REPUBLIC POLYTECHNIC

Lesson 16
Shortest Path and Maximum Flow Problems (II)

E211 – Operations Planning II

Learning Outcomes

At the end of the lesson, students should be able to:

- Identify a Shortest Path problem
- Apply Dijkstra's Algorithm to solve a Shortest Path problem
- Identify a Maximum Flow problem
- Apply Ford-Fulkerson's Algorithm to solve a Maximum Flow problem

Recap

Shortest Path Problem

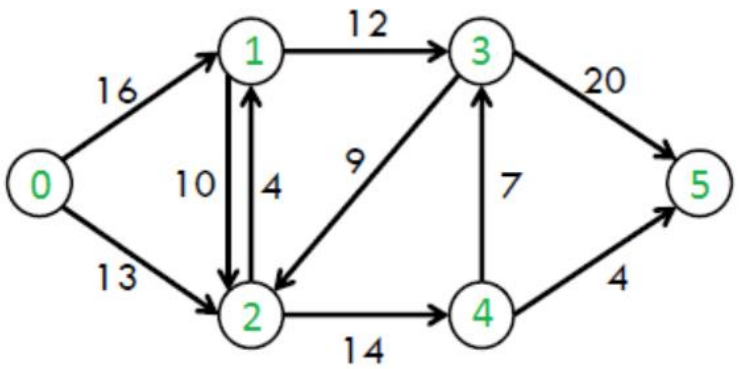
- The shortest path problem is to find the shortest distance, shortest time or minimum cost path between an origin and destination point(s).

- It is commonly found in logistics, route planning, network design, and many real-world problems where efficiency in movement or cost is essential.

- The key objective is to **minimize** the total distance, cost, or time required to reach a target point from a starting point.

Maximum Flow Problem

- The maximum flow problem is to find the maximum possible flow from an origin to a destination where the arcs of the network have limited flow capacities.
- Applications include supply chain optimization, design of fluid flow systems and communication network and transportation planning.
- The key objective is to **maximize** the amount of flow that can travel from the source to the sink, considering capacity limits on each path within the network
- Can you figure out that the Maximum Flow of the network (from 0 to 5) is 23?



Problem Statement

Scenario

- WantTheToys Pte Ltd is a Singapore based company that produces educational toys. Over the past 2 years, the company has expanded its production of various toys to cope with increasing demand from several countries

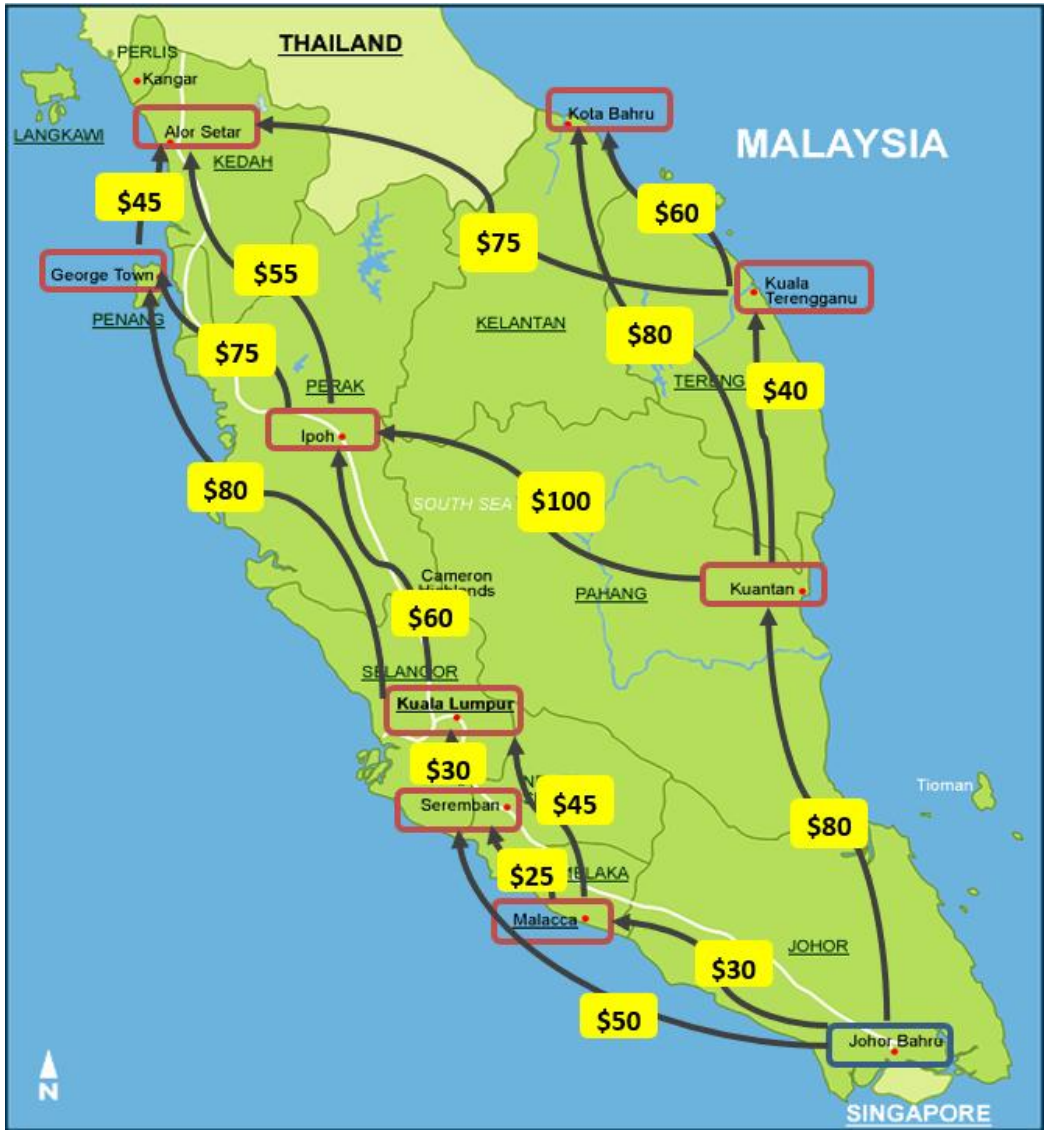
- Due to a new launch event in Malaysia, WantTheToys intends to ship additional units of its new toys from Singapore via Johor Bahru to major cities in Malaysia.

Shortest Path Problem

In the shortest path problem, the company wants to minimize the total cost of shipment from Johor Bahru to each of the other cities. What route should each shipment take?

Shown in the diagram are the shipment routes and cost of shipment per truckload.

What are some important assumptions when solving for the shortest path problem?



Maximum Flow Problem

In the maximum flow problem, there is limited shipping capacity along each route. WantTheToys would like to find out the maximum amount of the new toys it can ship from Johor Bahru to Alor Setar.

Shown in the diagram are shipment routes and route capacities.

What are some important assumptions when solving for the maximum flow problem?

Maximum Flow Problem (Part 2)

After the max flow is determined, there is a request for the company to deliver an additional 25 truckloads of toys from Johor Bahru to Alor Setar.

Based on the delivery routes planned out, suggest which routes' capacity should be expanded and the amount of flow to increase.

E-Learning Task

1. Complete the questions in ‘AY 2025 Sem 2 E211 L16 WS’. You may discuss in your team.

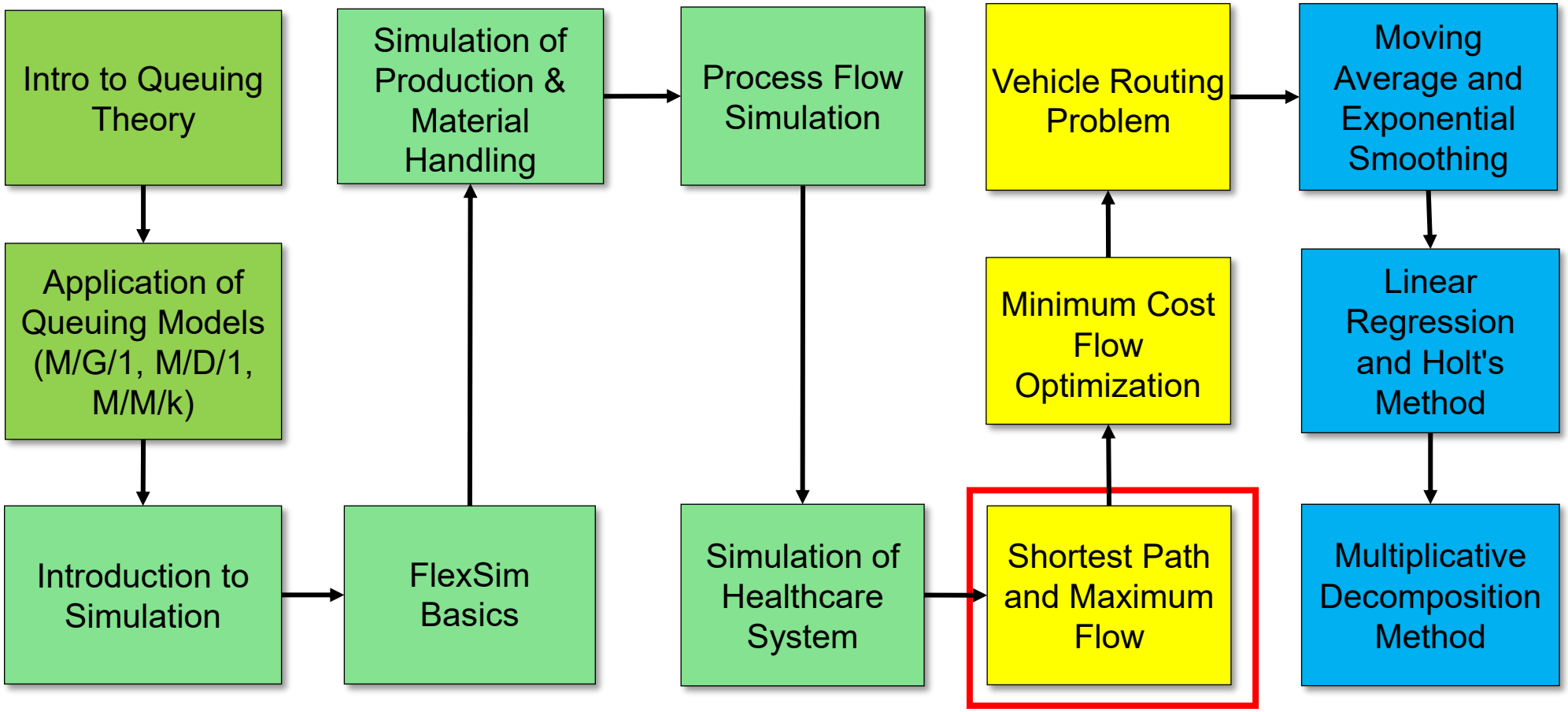
2. Attempt the SCORM quiz in the next page.

Remember to upload the completed WS to POLITEMall Student Submissions folder.

Quiz

Click the **Quiz** button to edit this object

In WantTheToys shortest path problem, how many ways are there to ship the toys from Johor Bahru to Ipoh?

- 2
- 3
- 4
- 5

E211 Progression